

# AI Agent for Information Retrieval: Generating and Ranking

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## Abstract

The field of information retrieval has significantly transformed with the integration of AI technologies. AI agents, especially those leveraging LLMs and vast computational power, have revolutionized information retrieval, processing, and presentation. LLM agents, with advanced memory, reasoning, and planning capabilities, can perform complex tasks, engage in coherent conversations, and provide personalized responses. Despite these advancements, challenges such as ensuring relevance and accuracy, mitigating biases, providing real-time responses, and maintaining data security remain. This workshop aims to explore these challenges, share innovative solutions, and discuss future directions. It will provide a platform to bring together researchers, practitioners to discuss the latest theoretical advancements and practical implementations of AI agents in information retrieval. Topics include AI in search, recommendation, and personalization systems. By gathering a diverse group of experts, the workshop seeks to deepen the understanding of AI agents in information retrieval, advance the field, and enhance its societal impact. Participants will gain insights into cutting-edge research, emerging trends, and foster knowledge exchange and collaboration within the community.

## CCS Concepts

• **Information systems** → **Information retrieval query processing; Retrieval models and ranking; Search engine architectures and scalability; Users and interactive retrieval; Evaluation of retrieval results; Document representation; Specialized information retrieval; Information retrieval query processing; Retrieval models and ranking; Search engine architectures and scalability; Users and interactive retrieval.**

## Keywords

Information Retrieval; AI agent; Large Language models (LLMs); Recommender Systems

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## 1 Workshop Themes and Topics

The advent of Large Language Model (LLM) Agents marks a significant milestone in AI and NLP. LLM Agents use large language models to perform tasks beyond text generation, such as conducting conversations, completing tasks, reasoning, planning, and demonstrating some degree of autonomous behavior. According to a recent market report [1], the LLM Agent market is estimated at USD 4.35 billion in 2023, with projections reaching USD 36.1 billion by 2030. The workshop aims to bridge the gap between information retrieval and AI technologies. It will explore the potential of generative models and AI-generated content to diversify available information and meet varied user preferences and needs. A key focus will be on the transformative potential of AI agents in reshaping user interactions with information retrieval systems, leveraging advancements in LLMs. The workshop will also address critical issues of trust and ethics in AI-driven information retrieval. Topic of Interests include but not limited to:

- Agentic Retrieval System
- Agentic Conversational Recommendation
- AI Agent for Personalization
- AI Agent for Sequential Recommendation
- AI Agent for Contextual Information Retrieval
- AI Agents for Cross-lingual and Multimodal Retrieval
- Retrieval-Augmented Generation Searching System
- Optimization of AI Agent Retrieval Models
- Bias Mitigation in AI-driven Information Retrieval
- Interpretable Generative Retrieval System
- Adversarial Attack and Defense in Agentic Retrieval System
- Ethical Considerations in Agentic Information Retrieval
- Human-AI Collaboration in Information Retrieval
- Evaluation for Agentic Information Retrieval
- Scalability and Efficiency in AI Agent Retrieval

## 2 Motivation, Goal and expected outcomes

Although research on LLM agent for information retrieval has progressed significantly, more challenges still need to be addressed. For instance, one of the key issues is enhancing the relevance and accuracy of retrieved information. Despite significant advancements,

ensuring that AI agents can consistently provide precise and contextually appropriate results remains a challenge. This workshop address those critical challenges, share innovative solutions, and explore future directions in the field. Our goal is to create a collaborative platform for researchers, practitioners, and industry experts to discuss the latest theoretical advancements and practical applications of AI agents in information retrieval. The workshop is expected to attract submissions from both academia and industry, encompassing various disciplines such as data science, machine learning, and etc. Participants will gain insights into cutting-edge research, emerging trends, and innovative solutions, fostering a vibrant community dedicated to advancing AI-driven information retrieval.

Encourage physical attendance and in-person activities:

- Invite high-profile and famous senior academics or industry leaders to be keynote speakers
- Require all accepted workshop papers to be presented in the workshop in person
- Set up a Best Paper Award in the workshop and announce the winner in the workshop.

### 3 Target Audience

The rapid advancements in AI-driven information retrieval, particularly with AI agents and large language models (LLMs), have captured significant interest from both research and industry communities. Addressing challenges such as improving information relevance and accuracy, managing vast data volumes, and ensuring real-time processing is crucial. This workshop aims to attract researchers and practitioners from academia and industry interested in the latest AI trends in information retrieval. With a focus on generative AI agents in search and recommendation, the workshop expects high participation, with 50 to 80 attendees. It aims to facilitate knowledge exchange and foster collaboration among experts dedicated to advancing AI-driven information retrieval technologies. Participants will benefit from discussions on cutting-edge research, innovative applications, and practical challenges in deploying AI agents for information retrieval, including areas like personalization, cross-lingual retrieval, and interpretability.

### 4 Workshop Relevance

CIKM is a premier conference for Information and Knowledge Management research. This workshop aligns perfectly with CIKM's mission. As information and knowledge management evolve, integrating AI agents into information retrieval systems becomes crucial. The workshop addresses fundamental challenges and explores innovative solutions at the intersection of AI technologies and information retrieval. Focusing on generating and ranking information, the workshop delves into theoretical advancements and practical applications of AI agents, including large language models (LLMs). By emphasizing AI-driven approaches and Agentic Systems, the workshop contributes to shaping the future landscape of information retrieval, making it a valuable addition to the CIKM conference.

## 5 Past Workshops

Relevant workshops run by some of us were held at KDD 2023-2021, WSDM 2023-2022, and CIKM 2022.

- Foundation and Applications in Large-scale AI models workshop was held at KDD 2023. It was one of the popular among all the workshops, with more than 100 attendees on-site. We had 4 highly influential keynote speakers from Google, Salesforce and IBM. We had around 30 submissions and 9 papers were accepted for presentation.
- Deep Learning for Search Recommendation workshop is a full-day workshop in conjunction with CIKM 2022. The workshop invited 5 keynote speakers from Google, Amazon etc. There are 27 submissions, among which 12 papers were accepted.
- The 1st, 2nd and 3rd International Workshop on Interactive and scalable information retrieval methods for e-commerce was held in conjunction with WSDM 2022, WWW 2023 and WSDM 2024. Invited speakers and submissions were from academia and industry, such as Nvidia, Amazon Search.
- The 2nd International Workshop on Industrial Recommendation Systems was held as a full-day workshop in conjunction with KDD 2021. There were 23 paper submissions, among which 11 papers were accepted. The workshop invited keynote speakers from Amazon, Netflix, Spotify, Nvidia.

## 6 Organizer Profiles

- **A/Prof. Yongfeng Zhang** is an Associate Professor in the Department of Computer Science at Rutgers University. His research interest is in Machine Learning and Data Mining, Information Retrieval and Recommender Systems, Natural Language Processing, Trustworthy AI and AI for Science. He regularly serves as Senior PC Member or Area Chair for SIGIR, WWW, KDD, ICLR, RecSys, ACL, CIKM, WSDM, AAAI, etc. He is a Distinguished Editor for ACM Transactions on Information Systems (TOIS), and also serves as Associate Editor for ACM Transactions on Recommender Systems (TORS) as well as Frontiers in Big Data. His research won the 2024 ACM SIGIR Test of Time Award. He is a Siebel Scholar of the class 2015 and an NSF career awardee in 2021. He has extensive experience in organizing workshops on top-tier conferences such as the 1st to 3rd Workshop on Explainable Recommendation and Search at SIGIR, and Workshop on Machine Reasoning at WSDM.
- **Dr. Zhiwei Liu** Dr. Zhiwei Liu is a senior research scientist at Salesforce AI Research. He received his Ph.D. degree from University of Illinois at Chicago. Dr. Liu's research interests lie in AI agents, Large Models, recommender systems, and data mining. He serves as the Guest Editor for ACM Transactions on Recommender System (TORS) and PC members for CIKM, KDD, WWW, SIGIR, WSDM, AAAI, RecSys, ICLR, ACL, EMNLP, NeurIPS, and etc. He has published over 40 top-ranking AI conference and journal papers, including CIKM, SIGIR, WWW, ICLR, EMNLP, WSDM, RecSys, ACM TIST.
- **Dr. Qingsong Wen** is the Head of AI Research & Chief Scientist at Squirrel Ai Learning Inc. Before that, he worked at

Alibaba, Qualcomm, and Marvell, and received his Ph.D. degrees in Electrical and Computer Engineering from Georgia Institute of Technology, USA. His research interests include machine learning and decision intelligence, especially AI for Time Series (AI4TS) & AI for Education (AI4EDU). He has published over 100 top-ranked AI conference and journal papers, had multiple Oral/Spotlight Papers at NeurIPS, ICML, and ICLR, had multiple Most Influential Papers at IJCAI, received multiple IAAI Deployed Application Awards at AAI, and won First Place of SP Grand Challenge at ICASSP. Currently, he serves as Organizer/Co-Chair of Workshop on AI for Time Series (AI4TS @ KDD, ICDM, SDM, AAI, IJCAI) and Workshop on AI for Education (AI4EDU @ CAI). He also serves as Associate Editor for Neurocomputing, Associate Editor for IEEE Signal Processing Letters, Guest Editor for Applied Energy, and Guest Editor for IEEE Internet of Things Journal. In addition, he has regularly served as Area Chair/PC of the AI conferences including NeurIPS, KDD, AAI, IJCAI, ICDM, ICASSP, etc.

- **Dr. Linsey Pang** is a Principal Machine Learning Scientist at Salesforce, working on deep learning recommendation modeling projects. Prior to this, she was a Principal Data Scientist at Walmart Lab. She was leading the Pricing, One demand data science projects. Prior to joining Walmart Lab, she worked as an applied scientist at eBay Inc. She finished her PhD degree in machine learning research at the University of Sydney. She co-organizes multiple workshops at CIKM, KDD, and WSDM. She also serves as SPC and PC for multiple top conferences.
- **A/Prof. Wei Liu** is an Associate Professor in Machine Learning, and the Director of Future Intelligence Research Lab, in the School of Computer Science, the University of Technology Sydney (UTS). He obtained his PhD degree in Machine Learning research at the University of Sydney. His current research focuses are adversarial machine learning, cybersecurity, game theory, multimodal machine learning, natural language processing, and intrusion detection. Wei has pub-

lished over 100 papers in CORE A\*/A and Q1 (i.e., top-prestigious) journals and conferences. He has received 3 Best Paper Awards. In addition, one of his first-authored papers received the Most Influential Paper Award in conference PAKDD 2021. Wei served as a tutorial chair at ICDM 2021. He is a co-organizer of the NeuroRec 2021 workshop collocated with ICDM 2021, the AdKDD 2022 workshop collocated with KDD 2022, and the DL4SR 2022 workshop collocated with CIKM 2022.

- **Prof. Philip S. Yu** is a Distinguished Professor in Computer Science at the University of Illinois at Chicago and also holds the Wexler Chair in Information Technology. Before joining UIC, Dr. Yu was with IBM at the Watson Research Center. His research interest is on big data, and artificial intelligence. He has published more than 1,600 papers in refereed journals and conferences. He holds or has applied for more than 300 US patents. Dr. Yu is a Fellow of the ACM and the IEEE. He received the VLDB 2022 Test of Time Award, ACM SIGSPATIAL 2021 10-year Impact Award, WSDM 2020 Honorable Mentions of the Test of Time Award, ICDM 2013 10-year Highest-Impact Paper Award, and the EDBT Test of Time Award (2014). He was the Editor-in-Chiefs of ACM Transactions on Knowledge Discovery from Data and IEEE Transactions on Knowledge and Data Engineering. Dr. Yu is also the recipient of ACM SIGKDD 2016 Innovation Award for his influential research and scientific contributions on mining, fusion and anonymization of big data, the IEEE Computer Society's 2013 Technical Achievement Award for *pioneering and fundamentally innovative contributions to the scalable indexing, querying, searching, mining and anonymization of big data*, and the Research Contributions Award from IEEE Intl. Conference on Data Mining (ICDM) in 2003 for his pioneering contributions to the field of data mining.

## References

- [1] M. Kumarashanmugam. 2024. *The Current State, Progress, and Future of LLM Agents*. <https://www.linkedin.com/pulse/current-state-progress-markets-future-llm-agents-kumarashanmugam-mj8wf/> Accessed: 2024-06-09.